Multi-Agent Scientific Paper Review

Powered by Phi-3 Mini · May 03, 2026 20:21

File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	BALANCED
Model	qwen3:8b

■ *AI-ASSISTED REVIEW NOTICE: This report was generated by an artificial intelligence system using Phi-3 Mini as the language model. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.*

Editorial Decision

MINOR REVISION

Weighted Score: 3.6 / 5.0
Confidence: 82%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**5.0**	Excellent	■■■■■
Manuscript Structure	**2.5**	Weak	■■■■■
Methodology & Equations	**4.0**	Good	■■■■■
Statistical Analysis	**3.5**	Good	■■■■■
Results & Evidence	**4.0**	Good	■■■■■
Discussion & Conclusions	**3.5**	Good	■■■■■
Figures & Tables	**4.0**	Good	■■■■■
Scientific Writing	**3.5**	Good	■■■■■
References & Citations	**2.5**	Weak	■■■■■
Ethics & Transparency	**2.5**	Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **5.0 / 5.0** | Confidence: 95%

Strengths:

- [TITLE] is specific, informative, and includes key terms for discoverability.
- [ABSTRACT] follows a structured format and clearly states the research problem, methods, results, and conclusions.
- [KEYWORDS] include a mix of specific and general terms, and are relevant to the domain.

Minor Comments:

- [TITLE] could be slightly shortened for clarity without losing specificity.

Recommendations:

- [TITLE] Consider shortening the title slightly for improved readability while maintaining specificity.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear title and abstract
- Well-defined methodology section
- Experimental results presented with figures

Weaknesses:

- Missing literature review section
- Repetition of content in section_vi and section_vii
- Research problem and hypothesis not clearly stated
- Objectives not explicitly stated or matched with conclusions

Major Comments:

- The manuscript lacks a literature review section, which is essential for contextualizing the research problem and establishing the novelty of the work.
- The sections section_vi and section_vii are repetitive and should be consolidated or removed.
- The research problem and hypothesis are not clearly stated in the introduction.
- The objectives are not explicitly declared and do not align with the conclusions.

Minor Comments:

- The section map includes some non-standard sections (e.g., preamble, biography, keywords) that may need clarification.
- The abstract is missing or incomplete as it is not included in the provided text.

Recommendations:

- Include a literature review section to provide context and justify the research problem.
- Consolidate or remove the repetitive sections section_vi and section_vii.
- Clearly state the research problem and hypothesis in the introduction.
- Explicitly declare the objectives and ensure they are addressed in the conclusions.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: 90%

Strengths:

- Clear integration of physical constraints into the neural network framework
- Multi-objective optimization approach for parameter estimation
- Use of a five-story building prototype for validation
- Logical flow from problem definition to experimental validation

Weaknesses:

- Lack of detailed description of the neural network architecture and training process
- No mention of specific optimization algorithms used (e.g., genetic algorithm, particle swarm optimization)
- Insufficient detail on the experimental setup and data preprocessing

Major Comments:

- The manuscript should provide more detailed information on the neural network architecture and training process to ensure reproducibility.

Minor Comments:

- Clarify the specific optimization algorithms used in the multi-objective framework.

Recommendations:

- Include a detailed description of the neural network architecture and training procedure.
- Provide more information on the experimental setup, data preprocessing, and validation metrics.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- The manuscript reports multiple statistical validation elements, including independent runs, nonparametric tests, correction procedures, confidence intervals, and performance/dispersion metrics.

Weaknesses:

- Could not parse model response — raw output stored

Recommendations:

- Re-run this agent or check model output

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- Figures are self-explanatory and coherent with the text
- Tables have descriptive headers and are consistent with the text
- Equations are numbered sequentially and are coherent with the methods described

Weaknesses:

- The extracted text does not expose table bodies, so data consistency cannot be fully verified
- Some figure descriptions are incomplete (e.g., 'Fig. 3: Displacement comparison (a) x4. (b) x5.')
- The text does not provide full context for all figures and tables, which may affect clarity

Major Comments:

- The extracted text does not expose table bodies, so data consistency cannot be fully verified.

Minor Comments:

- Some figure descriptions are incomplete and may require clarification.
- The text does not provide full context for all figures and tables, which may affect clarity.

Recommendations:

- Ensure that all figures and tables are fully described and that their content is clearly explained in the text.
- Provide full table bodies for verification of data consistency and statistical significance.

ResultsReviewer

Results validity and empirical support

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- MSE, RMSE, CV, and hypervolume metrics are reported, which are relevant to the multi-objective optimization problem
- Baseline comparisons with GA and PSO are included, which is critical for evaluating the proposed EANN framework
- Pareto analysis and spread indicators are mentioned, which are appropriate for multi-objective optimization

Weaknesses:

- The results section lacks specific details on the statistical validation of performance differences, such as confidence intervals or p-values
- The description of the experimental setup and the exact nature of the 30 independent runs is not sufficiently detailed
- The comparison with baseline methods is not quantitatively detailed, such as the exact performance metrics used for comparison
- The section does not clearly state whether negative results (e.g., cases where the EANN underperformed) were reported or acknowledged

Major Comments:

- The results section should include more detailed statistical validation to support the significance of performance differences between methods.

Minor Comments:

- The description of the 30 independent runs and their impact on the results should be more clearly explained.

Recommendations:

- Include confidence intervals or p-values to support the statistical significance of the results.
- Clarify the exact metrics used for comparing the EANN framework with GA and PSO baselines.
- Report any negative results or cases where the EANN underperformed, to ensure transparency.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 90%

Strengths:

- The discussion clearly states the limitations of the study, such as the computational cost of EANN compared to GA and PSO.
- The conclusions are directly derived from the results and clearly state the contribution of the proposed framework.

Weaknesses:

- The discussion lacks a detailed comparison with prior work, focusing only on citations without actual comparison.
- The future work directions are not specified or justified in the conclusions section.

Major Comments:

- The discussion should include a more detailed comparison with prior work to contextualize the findings within the existing literature.

Minor Comments:

- The conclusions could benefit from more specific future work directions to guide further research.

Recommendations:

- Include a detailed comparison with prior work in the discussion to contextualize the findings.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical terminology
- Well-structured sections
- Use of figures for data presentation

Weaknesses:

- Overuse of passive voice
- Lack of definition for acronyms
- Vague quantifiers without numerical support

Major Comments:

- The manuscript contains overused passive voice, which reduces clarity and engagement.
- Acronyms such as EANN, GA, and PSO are used without definition, which may confuse readers.
- The phrase 'significant' is used without numerical support, weakening the claim.

Minor Comments:

- Some sentences are overly long and could be split for better readability.
- The introduction lacks a clear research question or hypothesis.

Recommendations:

- Define all acronyms on first use (e.g., EANN = Evolutionary Artificial Neural Network).
- Replace vague quantifiers like 'significant' and 'many' with specific numerical values.
- Revise passive voice constructions to active voice where possible to improve clarity and engagement.
- Split long sentences into shorter, more digestible ones.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Recent references from 2021-2022
- Citations to relevant domains like magnetorheological dampers and system identification

Weaknesses:

- Lack of seminal foundational works
- Missing key areas in literature review
- Some references are weak or irrelevant

Major Comments:

- The references section lacks seminal foundational works that are critical for understanding the theoretical basis of the research.
- Key areas such as structural health monitoring and seismic design regulations are missing, which weakens the literature review.
- The reference [22] appears to have a typographical error in the author names and may be a weak or irrelevant citation.

Minor Comments:

- The citation style is consistent, but more attention to formatting could improve clarity.

Recommendations:

• Include seminal foundational papers in structural dynamics and control theory.

• Add references to key works in structural health monitoring and seismic design regulations.

• Correct the citation [22] to ensure accuracy and relevance.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

• Limitations are explicitly stated in the discussion, though they lack specificity.

• The study presents a novel framework for structural parameter identification, which is a significant contribution.

Weaknesses:

• Limitations are not specific or detailed, which reduces their credibility and utility.

• The manuscript lacks information on reproducibility, such as data and code availability.

• The absence of ethical approval, conflict of interest, funding, and data availability statements undermines the transparency and rigor of the research.

Major Comments:

• The manuscript fails to report essential ethical and transparency information, which is critical for publication in Q1 journals.

• Limitations are mentioned but not sufficiently detailed or specific, which weakens the credibility of the study.

Minor Comments:

• The methodology and results are well-presented, but the lack of reproducibility information is a minor concern.

Recommendations:

• Include a statement on ethical approval for any human or animal studies.

• Provide a data availability statement and ensure code and data are accessible.

• Declare any potential conflicts of interest and disclose funding sources.

• Elaborate on the limitations with specific examples and their impact on the conclusions.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] The manuscript lacks a literature review section, which is essential for contextualizing the research problem and establishing the novelty of the work.

[2] The sections section_vi and section_vii are repetitive and should be consolidated or removed.

[3] The research problem and hypothesis are not clearly stated in the introduction.

[4] The objectives are not explicitly declared and do not align with the conclusions.

[5] The manuscript should provide more detailed information on the neural network architecture and training process to ensure reproducibility.

[6] The extracted text does not expose table bodies, so data consistency cannot be fully verified.

[7] The results section should include more detailed statistical validation to support the significance of performance differences between methods.

[8] The discussion should include a more detailed comparison with prior work to contextualize the findings within the existing literature.

[9] The manuscript contains overused passive voice, which reduces clarity and engagement.

[10] Acronyms such as EANN, GA, and PSO are used without definition, which may confuse readers.

[11] The phrase 'significant' is used without numerical support, weakening the claim.

[12] The references section lacks seminal foundational works that are critical for understanding the theoretical basis of the research.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] [TITLE] could be slightly shortened for clarity without losing specificity.

[2] The section map includes some non-standard sections (e.g., preamble, biography, keywords) that may need clarification.

[3] The abstract is missing or incomplete as it is not included in the provided text.

[4] Clarify the specific optimization algorithms used in the multi-objective framework.

[5] Some figure descriptions are incomplete and may require clarification.

[6] The text does not provide full context for all figures and tables, which may affect clarity.

[7] The description of the 30 independent runs and their impact on the results should be more clearly explained.

[8] The conclusions could benefit from more specific future work directions to guide further research.

[9] Some sentences are overly long and could be split for better readability.

[10] The introduction lacks a clear research question or hypothesis.

[11] The citation style is consistent, but more attention to formatting could improve clarity.

[12] The methodology and results are well-presented, but the lack of reproducibility information is a minor concern.

Specific Improvement Recommendations

1. [TITLE] Consider shortening the title slightly for improved readability while maintaining specificity.
2. Include a literature review section to provide context and justify the research problem.
3. Consolidate or remove the repetitive sections section_vi and section_vii.
4. Clearly state the research problem and hypothesis in the introduction.
5. Explicitly declare the objectives and ensure they are addressed in the conclusions.
6. Include a detailed description of the neural network architecture and training procedure.
7. Provide more information on the experimental setup, data preprocessing, and validation metrics.

8. Re-run this agent or check model output
9. Ensure that all figures and tables are fully described and that their content is clearly explained in the text.
10. Provide full table bodies for verification of data consistency and statistical significance.
11. Include confidence intervals or p-values to support the statistical significance of the results.
12. Clarify the exact metrics used for comparing the EANN framework with GA and PSO baselines.
